

Ashwin Sateesh Kumar

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Machine Learning Engineer with 5+ years of experience building and deploying production AI systems spanning LLMs, agentic AI, multimodal, and computer vision — combining applied research with a recent focus on AI security.

EXPERIENCE

AI Software Engineer

Dec. 2024 – Present

Citibank - Chief Information Security Office

Dallas, TX

- Engineered a ReAct-based multi-agent vulnerability detection system with dependency-aware code analysis using LangGraph and AST parsing, improving detection accuracy by 25%
- Architected multi-agent infrastructure on OpenShift — FastAPI and LiteLLM gateway with circuit breakers and fallback chain, PostgreSQL + Redis state management, and HPA auto-scaling, serving 8,000+ analysts
- Designed LangGraph two-loop orchestration — supervisor LLM routing between parallel sub-agents with scoped MCP tools, mid-request checkpointing, and Tekton CI/Harness CD canary deployment
- Delivered a containerized browser automation agent using Stagehand with LLM gateway, enabling autonomous authenticated web operations for vulnerability research, cutting manual extraction from hours to <5 minutes
- Optimized a RAG pipeline with BM25 + cross-encoder reranking and built OTel/Prometheus observability with DeepEval evaluation on production traces, improving retrieval precision by 25% and faithfulness above 0.95

Machine Learning Engineer

Jan. 2024 – Dec. 2024

Abecedarian

Boston, MA

- Developed and deployed a multimodal AI assistant (text + image), fine-tuning Llama 3.1 8B with QLoRA and aligning it via RLHF (DPO) on human-ranked preference pairs — reaching a 72% win-rate over the SFT baseline
- Cut the assistant's image-generation latency 2× by distilling Stable Diffusion into a lightweight latent diffusion model via U-Net pruning, preserving CLIP alignment for real-time use

Research Assistant

July 2023 – Nov. 2023

Northeastern University

Boston, MA

- Uncovered demographic bias (visual and linguistic) in 400k social media posts by training a disentangled multimodal β -VAE with a Mixture of Experts (MoE) fusion layer over U-Net and BERT encoders
- Reduced inference latency by 30% on GCP via bfloat16 quantization, enabling counterfactual fairness auditing by manipulating disentangled latent factors across demographic groups

Machine Learning Engineer Intern, R&D

June 2022 – Dec. 2022

Signify Research (Philips Lighting)

Boston, MA

- Engineered a real-time user re-identification system using omni-scale feature learning, achieving 0.95 mAP in identity matching across multi-camera views, deployed for enterprise smart lighting optimization

Software Engineer

July 2019 – Nov. 2020

KPIT Technologies Ltd.

Bengaluru, India

- Developed an autonomous driving perception system using U-Net scene segmentation and vision-radar sensor fusion, achieving 0.89 IoU on real-time traffic object detection

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, R

Frameworks: PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, FastAPI, LiteLLM, React

Infrastructure: Git, Bash, Docker, Kubernetes, OpenShift, Tekton, Harness, n8n, Redis, PostgreSQL, Grafana

Cloud: GCP (Vertex AI, Cloud Storage), AWS (S3, SageMaker)

Libraries: NumPy, pandas, SciPy, OpenCV, NLTK, XGBoost, FAISS, HF Transformers, RAGAS, OpenTelemetry

PROJECTS

Agricultural Policy Recommendation System

Python, PyTorch, Earth Engine

- Recommended state-level agricultural policies by forecasting crop yield with 5% MAPE from satellite imagery (ViT + XGBoost), demand-supply data (LSTM), and news data (BERT)
- Orchestrated through LangChain and a fine-tuned GPT-3.5 that turned multivariate forecasts into actionable policies, served via a Streamlit interface

EDUCATION

Northeastern University

Boston, MA

Master of Science in Data Science (GPA: 3.8/4.0)

Sep. 2021 – Dec. 2023

PES University

Bengaluru, India

Bachelor of Engineering in Electronics and Communications

Aug. 2015 – May 2019